

Problem

A radio receiver has an input resistance of $300\ \Omega$. When it is connected directly to an antenna system with a characteristic impedance of $75\ \Omega$, an impedance mismatch occurs. By inserting an impedance-matching transformer ahead of the receiver, maximum power can be realized. Calculate the required turns ratio.

Step-by-step solution

Step 1 of 1

$$Z_{TH} = \frac{Z_L}{n^2}$$

Or,

$$n = \sqrt{\frac{Z_L}{Z_{TH}}}$$

$$= \sqrt{\frac{75}{300}} = 0.5 .$$